화학반응속도론 (2014년 2학기)
제7장 액체상에서 분자의 운동과 화학반응

중요 학습 내용

- 분자의 확산과 확산식
- 확산식과 확산지배반응
- 전자전달반응과 반응메커니즘

확산과 Fick의 제1법칙

- 확산 (diffusion): 두 종류 이상의 기체 상 또는 액체 상 입자들이 서로 섞이는 과정 또는 현상
- 확산 계수 (diffusion coefficient): 입자들의 확산 정도를 알려주는 입자 고유의 값

Synoptic table 21.8* Diffusion coefficients at 298 K

	$D/(10^{-9}\ m^2\ s^{-1})$
H^+ in water	9.31
I_2 in hexane	4.05
Na^+ in water	1.33
Sucrose in water	0.522

* More values are given in the *Data section*.

- Fick의 제1법칙:

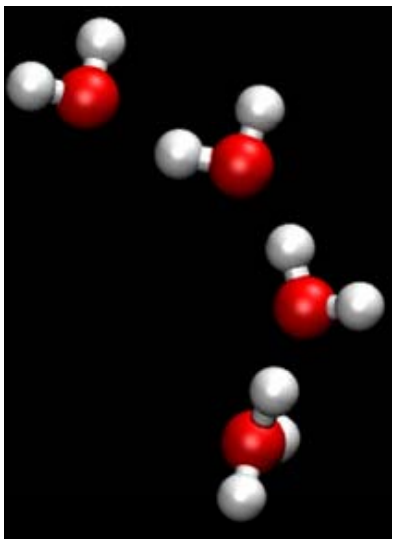
$$J(x,t) = -D\frac{dc(x,t)}{dx}$$



(예제) 세포막 외부와 내부의 **K^+** 이온의 농도 차는 일반적으로 **100 mM**이다 **(**내부가 외부보다 더 많다고 가정**).** 세포막 두께를 대략 **5 nm, K^+** 이온의 확산계수를 **2.0×10^{-5} cm^2/s**라고 할 때**, Fick** 제**1**법칙을 이용하여 **K^+** 이온의 **flux**를 계산하여라**.**

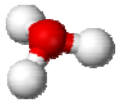
Proton transfer and Grotthuss mechanism

- Grotthuss mechanism

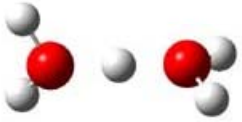


Marx, ChemPhysChem **7**, 1849 (2006)

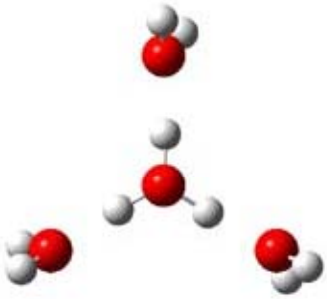
- The structure of hydrated proton
  - Hydronium ion



  - Zundel cation ($H_5O_2^+$)

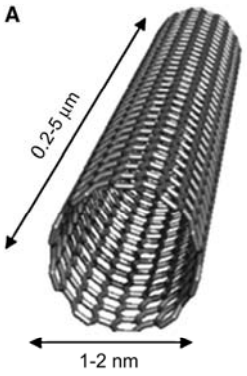


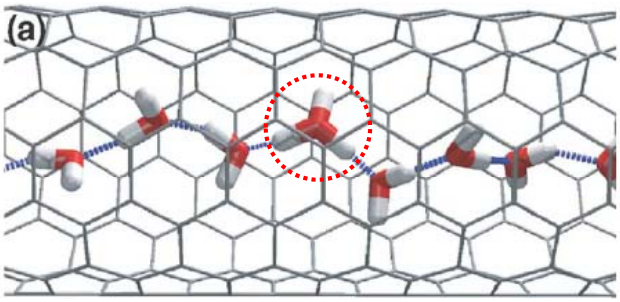
  - Eigen cation ($H_9O_4^+$)



Proton diffusion in carbon nanotubes

- Carbon nanotube





system	D_{H+} (Å^2/ps)
Bulk water (simulation)	0.4
Water in carbon nanotube (simulation)	3 - 17
Bulk water (experiment)	0.93

Dellago et al., Phys. Rev. Lett. **90**, 105902 (2003)

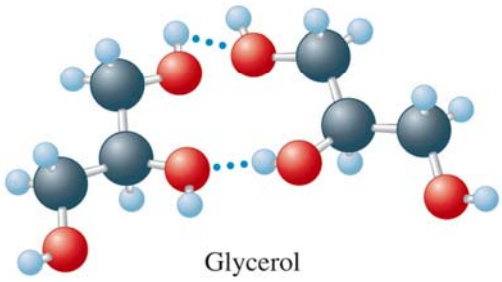
점성도

점성도 (viscosity): 흐름에 대한 유체 저항의 척도.

물의 수소결합과 글리세롤의 수소결합은 점성도를 증가시킨다.

TABLE 11.3 Viscosity of Some Common Liquids at 20°C

Liquid	Viscosity (N s/m^2)*
Acetone (C_3H_6O)	3.16×10^{-4}
Benzene (C_6H_6)	6.25×10^{-4}
Blood	4×10^{-3}
Carbon tetrachloride (CCl_4)	9.69×10^{-4}
Ethanol (C_2H_5OH)	1.20×10^{-3}
Diethyl ether ($C_2H_5OC_2H_5$)	2.33×10^{-4}
Glycerol ($C_3H_8O_3$)	1.49
Mercury (Hg)	1.55×10^{-3}
Water (H_2O)	1.01×10^{-3}


Glycerol





Stokes-Einstein 식과 이온이동도

- Stokes-Einstein 식

$$D = \frac{k_B T}{6\pi\eta R_h}$$ (η: 용매의 점성도, R_h: 용질의 hydrodynamic radius)

- 이온이동도 (ionic mobility)

$$u = \frac{ze}{k_B T} D$$

Table 20.5* Ionic mobilities in water at 298 K

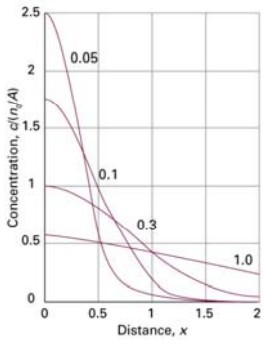
	$u/(10^{-8}\ m^2\ s^{-1}\ V^{-1})$		$u/(10^{-8}\ m^2\ s^{-1}\ V^{-1})$
H^+	36.23	OH^-	20.64
Na^+	5.19	Cl^-	7.91
K^+	7.62	Br^-	8.09
Zn^{2+}	5.47	SO_4^{2-}	8.29

* More values are given in the *Data section*.

Fick의 제2법칙과 Einstein-Smoluchowski 식

- Fick의 제2법칙과 확산식

$$\frac{\partial c(x,t)}{\partial t} = D\frac{\partial^2 c(x,t)}{\partial x^2}$$



- 확산식의 일반해

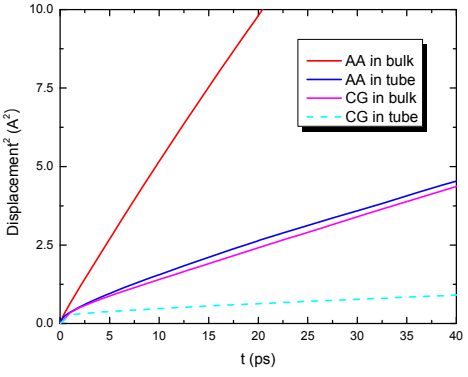
$$c(x,t) = \frac{n_0}{A(4\pi Dt)^{1/2}} e^{-(x-x_0)^2/4Dt}$$

- 확산확률

$$P(x,t) = \frac{1}{(4\pi Dt)^{1/2}} e^{-(x-x_0)^2/4Dt}$$

- 평균제곱변위와Einstein-Smoluchowski 식

$$\langle x^2 \rangle = 2Dt$$



용액에서의 반응 메커니즘

- 용액에서의 반응

$$\mathrm{A} + \mathrm{B} \underset{k_{-d}}{\overset{k_d}{\rightleftharpoons}} \mathrm{AB} \xrightarrow{k_a} \mathrm{P}$$

- 정류상태근사법

$$\frac{d[\mathrm{AB}]}{dt} = k_d[\mathrm{A}][\mathrm{B}] - k_{-d}[\mathrm{AB}] - k_a[\mathrm{AB}] \approx 0 \Rightarrow [\mathrm{AB}]_{\mathrm{ss}} = \frac{k_d}{k_a + k_{-d}}$$

$$\frac{d[\mathrm{P}]}{dt} = k_a[\mathrm{AB}] = \frac{k_a k_d}{k_a + k_{-d}}[\mathrm{A}][\mathrm{B}] = k_{\mathrm{obs}}[\mathrm{A}][\mathrm{B}]$$

$$k_{\mathrm{obs}} = \frac{k_a k_d}{k_a + k_{-d}}$$

Diffusion-controlled reaction

- 확산지배반응 (diffusion-controlled reaction) ($k_a >> k_d$)

$$k_{\mathrm{obs}} = \frac{k_a k_d}{k_a + k_{-d}} \approx k_a$$

- 활성화지배반응 (activation-controlled reaction) ($k_a << k_{-d}$)

$$k_{\mathrm{obs}} = \frac{k_a k_d}{k_a + k_{-d}} \approx k_a \frac{k_d}{k_{-d}}$$

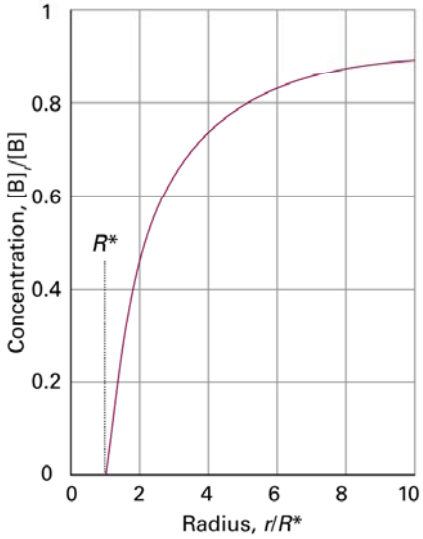
Smoluchowski 식과 정류상태근사해

- Smoluchowski 식

$$\frac{\partial c(\mathbf{r},t)}{\partial t} = D\nabla^2 c(\mathbf{r},t)$$

- 정류상태 근사해

$$\frac{d^2c(r,t)}{dr^2} + \frac{2}{r}\frac{dc(r,t)}{dr} = 0$$



- 속도법칙과 속도상수 k_d

$$\frac{dc(R^*)}{dt} = 4\pi R^{*2} J(R^*) = 4\pi R^* Dc(R^*) \Rightarrow k_d = 4\pi R^* D$$

7.3 균일계에서 전자전달반응

- 전자전달반응의 속도법칙

$$\mathrm{D} + \mathrm{A} \xrightarrow{k_{\mathrm{obs}}} \mathrm{D}^+ + \mathrm{A}^- \Rightarrow v = k_{\mathrm{obs}}[\mathrm{D}][\mathrm{A}] = k_{\mathrm{d}}[\mathrm{D}^+][\mathrm{A}^-]$$

- 전자전달반응의 소반응들

$$\mathrm{D} + \mathrm{A} \underset{k_{-\mathrm{a}}}{\overset{k_{\mathrm{a}}}{\rightleftharpoons}} \mathrm{DA}$$

$$\mathrm{DA} \underset{k_{-\mathrm{et}}}{\overset{k_{\mathrm{et}}}{\rightleftharpoons}} \mathrm{D}^+\mathrm{A}^-$$

$$\mathrm{D}^+\mathrm{A}^- \xrightarrow{k_{\mathrm{d}}} \mathrm{D}^+ + \mathrm{A}^-$$

$$\frac{1}{k_{\mathrm{obs}}} = \frac{1}{k_a} + \frac{k_{-\mathrm{a}}}{k_{\mathrm{a}}k_{\mathrm{et}}}\left(1 + \frac{k_{-\mathrm{et}}}{k_{\mathrm{d}}}\right)$$

전자전달반응의 확산지배반응과 활성화지배반응

- D^+A^-의 해리과정이 매우빠르다는 가정, 즉 $k_d >> k_{et}$라는 가정하에

$$\frac{1}{k_{\mathrm{obs}}} = \frac{1}{k_a} + \frac{k_{-\mathrm{a}}}{k_{\mathrm{a}} k_{\mathrm{et}}}\left(1 + \frac{k_{-\mathrm{et}}}{k_{\mathrm{d}}}\right) \approx \frac{1}{k_{\mathrm{a}}}\left(1 + \frac{k_{-\mathrm{a}}}{k_{\mathrm{et}}}\right)$$

  - 확산지배반응 ($k_{et} >> k_a$)

$$\frac{1}{k_{\mathrm{obs}}} \approx \frac{1}{k_{\mathrm{a}}} \Rightarrow k_{\mathrm{obs}} \approx k_{\mathrm{a}}$$

  - 활성화지배반응 ($k_{et} << k_a$)

$$\frac{1}{k_{\mathrm{obs}}} \approx \frac{k_{-\mathrm{a}}}{k_{\mathrm{a}} k_{\mathrm{et}}} \Rightarrow k_{\mathrm{obs}} \approx \frac{k_{\mathrm{a}}}{k_{-\mathrm{a}}} k_{\mathrm{et}}$$

전자전달반응과 Marcus 이론

- 전자전달반응에 대한 전이상태이론

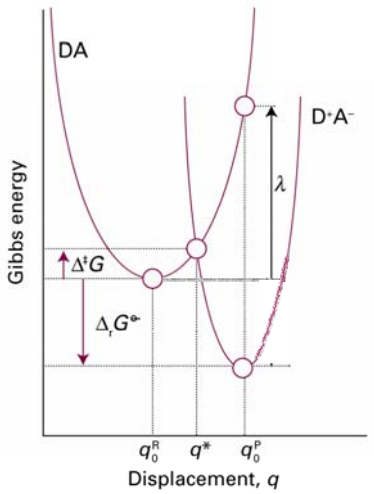
$$k_{\mathrm{et}} = \kappa \nu^{\ddagger} e^{-\Delta^{\ddagger} G / RT}$$

- 활성화에너지에 대한 Marcus 이론

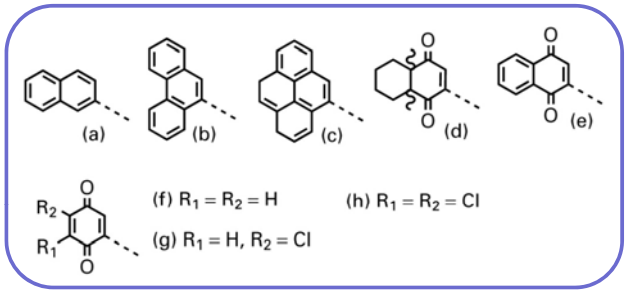
$$\Delta^{\ddagger} G = \frac{(\Delta_r G^{\circ} + \lambda)^2}{4\lambda}$$

- 터널 현상과 반응속도

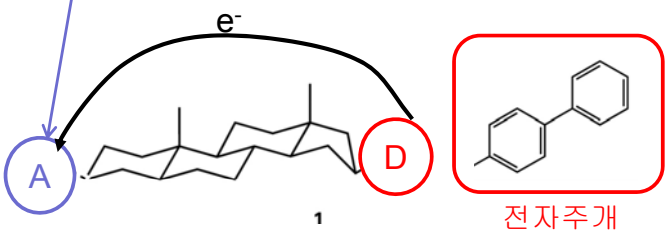
$$k_{\mathrm{et}} \propto e^{-\beta r} e^{-\Delta^{\ddagger} G / RT}$$



전자전달반응 실험 결과

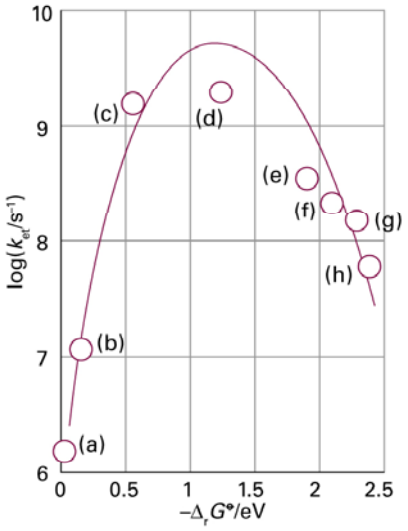


전자받개
(electron acceptor)



1

전자주개
(electron donor)



용매: 2-methyltetrahydrofuran
온도: 296 K

J. Am. Chem. Soc. **106**, 3047 (1984)